NLP4RE ID Card

The NLP4RE ID-Card is a means of capturing a descriptive overview of an NLP4RE paper by highlighting key information that can be useful in diverse scenarios such as replication, teaching support, or secondary research.

The ID-Card consists of seven dimensions concerning (I) the problem tackled by the paper (RE Task), (II) the solution proposed (NLP task(s)), (III) the input and output of the solution, (IV) the raw data and the annotated dataset, (V) the annotators and annotation process, (VI) the implementation of the solution proposed in the paper (Tool), and finally (VII) how the solution is evaluated (Evaluation).

To get an informative summary, the ID-Card can be applied on one RE task at a time. Multiple Cards can be used to describe a paper tackling multiple RE tasks (e.g., domain model generation and incompleteness detection). However, the ID-Card allows multiple answers in most of the questions, since the same RE task (e.g., requirements classification) can be solved using multiple NLP tasks (e.g., information extraction and classification).

In the following, we present some hints to facilitate generating an ID-Card for a particular paper:

- We included the field "Other/Comments" as a possible answer in all questions. This field can be used to add remarks when the question is not applicable, the answer is not specified in the paper, introduce another alternative that is not among the possible answers, or further comments about the question.

- In Dimension III (NLP Task Details), we are interested in the "initial input" and the "final output" of the proposed solution. Different questions about the output are adapted to suit the NLP task.

- In case a dimension or specific questions are not considered in the paper (e.g., the paper uses a dataset from the existing literature and so no details are given about the annotation process), then the dimension/questions can be skipped.

- If a question with a free-text field has multiple answers (the paper is using multiple datasets in IV), then the respective answers should be provided in a comma-separated format (e.g., "dataset1, dataset2, dataset3").

Title and authors of the paper

What About Emotions? Guiding Fine-Grained Emotion Extraction from Mobile App Reviews
Quim Motger, Marc Oriol, Max Tiessler, Xavier Franch, Jordi Marco

I. Re Task

I.1. What RE Task is your study addressing?
Please choose one option only, if multiple options apply fill multiple sheets

- Requirements classification
- Requirements tracing
- Requirements defect detection
- Model generation
- Test generation
- Requirements retrieval
- Information extraction from legal documents
- **App review analysis**
- Dependency and relation extraction
- Information extraction (e.g., features, terms) from requirements
- Other

Other/Comments:

II. NLP Task(s)

II.1. What types of NLP task is your study tackling?
Select all options that apply

- Classification (choose among classes)
- Translation (models, tests, etc.)
- **Information Extraction (glossary, terms)**
- Information Retrieval (search / rank)
- Other

Other/Comments:

III. NLP Task Details

Input Granularity

III.1. What is the input of your NLP task? *Select all options that apply*

- [ ] Document
- [ ] Paragraphs
- [x] Sentences
- [ ] Phrases
- [ ] Words
- [ ] Structured/tabular text: e.g., use case specification, feature table
- [ ] Other (e.g., models, trace links, diagrams, code comments)

Other/Comments:

Output Type (NLP Task: Classification)

Fill in only if you answered Classification in Section II

III.2. What type of classification is the study about? *Select all options that apply*

- [ ] Binary-Single Label (e.g., ambiguous vs unambiguous)
- [ ] Binary-Multi label (e.g., functional vs non-functional)
- [ ] Multi class-Single label (e.g., predicting number of stars, given a review. any scoring task)
- [x] Multi class- Multi label (e.g., type of quality in non-functional requirements, feature requests vs. bug reports vs. other)
- [ ] Other

Other/Comments:

III.3. What are the labels that can be assigned?
Please provide the labels, or point to the specific part of the original paper where the labels are specified

Joy, Sadness, Anticipation, Trust, Surprise, Disgust, Anger, Fear, Neutral

Output Type (NLP Task: Information Extraction)

Fill in only if you answered Information Extraction in Section II

III.4. What is the level of granularity of the extracted elements?
Select all options that apply

- [x] Sentences
- [ ] Phrases
- [ ] Words
- [ ] Other

Other/Comments:

III.5. What is the type of the extracted elements?

string (emotion)

Output Type (NLP Task: Translation)

Fill in only if you answered Translation in Section II

III.6. What is the type of output? *Select all options that apply*

- [ ] Text (e.g., from a diagram generate description)
- [ ] Table (e.g, from requirements generate a table of features)
- [ ] Graphical diagram (e.g., generate UML diagram from text)
- [ ] Executable model (e.g., from requirements to Simulink or executable sequence/activity/state diagrams)
- [ ] Test Cases
- [ ] Other

Other/Comments:

III.7. What is the translation mapping cardinality between initial input and final output? *Select all options that apply*

- [ ] 1 to 1
- [ ] 1 to many (e.g., 1 use case description to 3 diagrams)
- [ ] Many to 1 (e.g., many requirements translated into a single model, many feature table translated into a single feature diagram)
- [ ] Many to many (e.g., from many use case descriptions to one class diagram and multiple other diagrams)
- [ ] Other

Other/Comments:

Retrieved Elements (NLP Task: Information Retrieval)

Fill in only if you answered Information Retrieval in Section II

III.8. What are the types of elements retrieved with the query? *Select all options that apply*

- [ ] Document
- [ ] Paragraphs
- [ ] Sentences
- [ ] Phrases
- [ ] Words

Other/Comments:

Output Type (NLP Task: Other)

Fill in only if you answered Other in Section II

III.9. What is the type of output?

IV. Data and Dataset

IV.1. How many data items do you process?

Please report the numerical information and details about all the data that is used in your evaluation

1,112 unique sentences from app reviews, manually labeled by instructed annotators with 1,272 emotion labels, distributed as so: Joy (372), Sadness (303), Anticipation (182), Trust (151), Surprise (54), Disgust (49), Anger (27), Fear (24), Neutral (88), Reject (22).

IV.2. In which year or interval of years was the data produced?

Data collection of reviews was conducted in 2023. Data annotation was conducted in 2024.

IV.3. What is the source of the data?

Other/Comments:

- Industrial Project, proprietary data
- Industrial Project, publicly available data
- Community-based Open Source Projects (i.e., the community is organic and there is no top-down control from a company)
- Textbook examples or cases
- Student Projects (i.e., requirements developed by students in a realistic settings, where the project lasts months)
- Toy Requirements (e.g., used for courses, developed by students during exercises)
- Legal/regulatory documents
- **User generated content (e.g., app stores, user forums, ...)**
- Other

IV.4. What is the level of abstraction of the data (not limited to requirements)?

Other/Comments:

- **User-level (e.g., "Uploading pictures with the app is so annoying!")**
- Normative-level (e.g., "The processor shall not engage another processor without prior consent of the controller.")
- Business-level (e.g., "Provide customer with a pleasant online shopping experience")
- System-level (e.g., "When a GSI component constraint is violated, STS shall deliver a warning message to the system opera
- Module-level (e.g., module design description, sub-system design description)
- Code-level (e.g., code or comments)
- Other

Type of data and formulation

IV.5. What is the format of the data?

Please consider all the data that are used in the paper

Other/Comments:

- "Shall" requirements
- User stories
- Use cases
- **User reviews**
- Social media posts (e.g., Tweets)
- Bug/defect reports
- Messages in User forums

IV.6. How rigorous is the format of the data?

Other/Comments:

- **Unconstrained natural language**
- Template-based controlled natural language (i.e. the language follows a template or at least complies to some basic rules)
- Restricted grammar based controlled natural language (i.e., the language follows a restricted grammar)
- Semantically-augmented natural language (i.e., tags identifying semantic functions of part of the text, normally export from tools)
- Other

IV.7. What is the natural language of the data (if applicable)? English

Heterogeneity

IV.8. Please list which domains your data belongs to (e.g., automotive, satellite, entertainment, information systems) [comma separated text]

productivity, communication, tools, social, health and fitness, personalization, travel, maps and navigation, lifestyle, weather

IV.9. From how many different sources does your data come from?

Two (2): Google Play, AlternativeTo

Licensing

IV.10. Is the dataset publicly available (also from other authors)?

Fully

Other/Comments:

IV.11. What license has been used?

License: modification for any purposes

Other/Comments:

IV.12. Where is the dataset stored?

Other/Comments:

- On a private/corporate website
- **In a repository (GitHub, GitLab)**
- **In a persistent platform with DOI (Zenodo, FigShare)**
- Other

IV.13. Provide a URL to the dataset, if available, or to the original paper that proposed the dataset

https://github.com/nlp4se/review-emotion-analysis
https://doi.org/10.6084/m9.figshare.28548638

V. Annotators and Annotation process

If no annotation has been done, go to Section VI.

Refer to the annotations for the NLP task listed in Section II.
If you have multiple NLP tasks, separate your answers below with a comma.

V.1. How many annotators have been involved? 5

V.2. How are the entries annotated?

- [x] Multiple annotators per entry
- [ ] One annotator per entry (no quality control)
- [ ] One annotator per entry (quality control, possibly on a sample, by a supervisor)
- [ ] Partly multiple annotators per entry, partly one annotator per entry
- [ ] Other

Other/Comments:

V.3. What is the average level of application domain experience of the annotators?

- [ ] None or unknown
- [ ] Informed outsider
- [ ] Domain expert
- [x] Other

Other/Comments: Instructed annotators (researchers in the field of software engineering)

V.4. Who are the annotators?

- [x] The designers of the technique/tool
- [ ] People who have direct contact with the designers
- [ ] Independent annotators (e.g., crowd workers, pre-annotated data)
- [ ] Other

Other/Comments:

Annotation Scheme

V.5. How was the annotation scheme established among the annotators?

- [ ] Only via class labels
- [ ] Oral agreement among the annotators
- [ ] Written guidelines with label definitions
- [x] Written guidelines with definitions and examples
- [ ] Other

Other/Comments:

V.6. Did the authors make the written guidelines public?

- [x] Yes
- [ ] No
- [ ] Other

Other/Comments:

Annotation Process

V.7. Did the authors share other information that could support the annotators other than the elements to annotate?

- [ ] No
- [ ] Surrounding context (e.g., the entire paragraph where the sentence to annotate appears)
- [x] Entire document
- [ ] Other

Other/Comments:

V.8 Did the authors employ techniques to mitigate fatigue effects during the annotation sessions?

- [x] Yes
- [ ] No
- [ ] Other

Other/Comments:

Agreement

V.9. What are the metrics used to measure intercoder reliability?

- [x] Cohen's K
- [ ] Krippendorf's Alpha
- [ ] Fleiss' K
- [ ] Other

Other/Comments:

V.10. How were conflicts resolved?

- [x] Majority voting
- [x] Discussion among annotators
- [ ] Resolution by authors
- [ ] Resolution by independent expert (not an annotator)
- [ ] Disagreements were disregarded
- [ ] Not resolved
- [ ] Other

Other/Comments:

V.11. What is the measured agreement?

Average Cohen's Kappa: 0.69

VI. Tool

By tool, we intend any implementation of the approach(es) proposed in the paper. This includes scripts, executable programs, APIs, etc.

VI.1. What is the type of proposed solution?

- [ ] Supervised ML
- [ ] Unsupervised ML
- [ ] Supervised DL
- [ ] Unsupervised DL
- [ ] Rule-based
- [x] Other

Other/Comments:

Few-shot prompt-based method using LLMs

VI.2. What algorithms are used in the tool?

Few-shot prompt-based method using LLMs

VI.3. What has been released?

- [ ] Tool (e.g., a binary file) -- standalone
- [ ] Service on the web (e.g., REST)
- [ ] Library / API
- [x] Source code
- [ ] Executable notebook (e.g., Jupyter)
- [ ] Pre-trained model (ML-based trained model)
- [ ] No tool has been released
- [ ] Other

Other/Comments:

If no tool has been released go to Section VII.

VI.4. What needs to be done for running the tool?

- [ ] No installation is needed (e.g., standalone tool)
- [ ] Import and integrate into your own code (e.g., via MAVEN)
- [ ] Compile and run
- [ ] Virtual machine / Docker container
- [x] Reproduce the tool from the explanation in the pape
- [ ] Other

Other/Comments:

VI.5. What type of documentation has been provided alongside the tool?

- [x] README file
- [ ] Pseudocode / illustration in the paper
- [ ] Wiki or dedicated website
- [ ] Tutorial
- [x] Ready-to-use examples
- [x] An academic paper
- [ ] No documentation
- [ ] Other

Other/Comments:

VI.6. What type of dependencies does the tool have?

- [ ] None
- [x] Open source libraries / software (e.g., python libraries, NLTK)
- [ ] Proprietary libraries / software (e.g., Google Cloud NLP)
- [ ] Specific OS
- [ ] Specific hardware (e.g., hardware GPU acceleration required)
- [ ] External knowledge bases (e.g., Wikipedia)
- [ ] Other

Other/Comments:

VI.7. How is the tool released?

- [ ] On a private/corporate website
- [x] In a repository (GitHub, GitLab)
- [ ] In a persistent platform with DOI (Zenodo, FigShare)
- [ ] Other

Other/Comments:

VI.8. What license has been used?

- [ ] No license
- [ ] License: reuse only for non-commercial purposes
- [ ] License: reuse for any purposes
- [ ] License: modification only for non-commercial purposes
- [x] License: modification for any purposes
- [ ] Other

Other/Comments:

VI.9. Where is the tool released? [url]

https://github.com/nlp4se/review-emotion-analysis

VII. Evaluation

VII.1. What metrics are used to evaluate the approach(es)?

	Other/Comments:
Precision/Recall	
F-Score	
Accuracy	
AUC	
MAP	
LAG	
WER (word error rate)	
NIST - METEOR - ROUGE - BLEU	
Other	

VII.2. What is the validation procedure?

	Other/Comments:
Train-test split	
Cross-validation	
Cross-project validation	
Entire dataset (e.g., no training)	
Other	

VII.3. What baseline do you compare against?

	Other/Comments:
Existing tool or algorithm (e.g., from another paper for the same NLP task)	
Reconstructed tool from other research	
Automated, but self-defined (e.g., using ML algorithm not reported elsewhere)	
Theoretical / conceptual (e.g., majority class)	
Human baseline (i.e., comparison against human performance)	
None	
Other	

VII.4. Please provide more details about the baseline you compare against, if any.

Ground truth generated by human annotators.